Task: Product matching for a UK aesthetic-pharmacy price-comparison site

Context

`products_input.csv` (same folder) has 11,755 rows: `id, supplier, title, url`.
These come from 24 UK online pharmacies/wholesalers selling to aesthetic clinics and prescribers.
The same product appears many times under slightly different titles. Goal: group rows that are
truly the same product, using the PRODUCT PAGE and the OFFICIAL UK REFERENCE, not the title.
Prices are NOT needed now. Do not log in anywhere; many pages hide prices behind login, ignore that.

Core principle

The title is unreliable. Never decide route, strength, volume, pack size or form from the title alone.
Decide from (1) the supplier product page text, options and image, checked against (2) the original
UK presentation from the manufacturer / official source.

Pipeline

Step 1 - Scrape every url (cache everything, resumable)

- Use Python + requests/httpx, fall back to Playwright for JS pages. Run `python -I` for scripts that handle downloaded data.
- Save raw HTML to `cache/<supplier>/<id>.html`. Skip ids already cached (must be resumable).
- Polite: max 1 request/second per domain, retry with backoff, respect robots.txt, identify with a generic user agent.
- Extract per page: page title, full description/short description, attributes and variant/option selectors

(e.g. "IV / IM", "6 pack / 10 pack"), main product image URL, stock text, breadcrumb/category.

- Write `pages.jsonl`. Pages that fail go to `scrape_errors.csv` (do not guess for them).
- Many suppliers share a platform (WooCommerce, Shopify...). Write one extractor per platform, not per site. Domains: churchpharmacy, acrepharmacy, johnbannonpharmacy, twofaceaesthetics, lpgclinicswholesale, ask-pharmacy, healthxchange, dradamspharmacy, foxpharma, medfx, refinepharma, smilepharma, mypharmacyspace, wimpolefillers, hesupplies, skinonyou, aestheticpharma, aestheticsource, modapharmacy, kissaestheticswholesale, harpargrace, igoaesthetics, aestheticare, dermafocus.

Step 2 - Build the reference catalogue (once per product family, not per row)

For each brand/product family found, read the original UK source: emc (medicines.org.uk) SmPC/PIL, MHRA,
the manufacturer's UK site, IFU for devices. Record in `reference_catalog.csv`:
family, brand, manufacturer, UK pharmaceutical form, route, every UK presentation (strength, volume per unit,
pack contents, pack sizes), whether lidocaine versions exist, UK vs non-UK (Korean/EU) versions, source URLs.
Example: Pabrinex IV = pairs of 5 ml ampoules (No.1 + No.2), packs of 6 or 10 pairs (and a 10 ml presentation per the emc SmPC).
Do not invent presentations. If no official source is found, mark `reference_status = not_found`.

Step 3 - Classify each row from page + reference

Identity hierarchy:

- FAMILY = brand + product line (e.g. Pabrinex, Juvederm Ultra, Aqualyx).
- VARIANT = things that make it a different product: route (IV / IM / SC / ID / oral / topical), active strength or composition, lidocaine yes/no (fillers), UK vs Korean/EU version, form (tablet vs injection).
- PACK = same product, different count/packaging: number of pairs/vials/syringes, ampoule volume, box vs single.
  Ampoule volume and pack count are NEVER variant identity.

Rules (from the domain owner - follow strictly)

1. Read the product's original form first: tablet, liquid, IV injection, IM injection, etc. Then read the supplier description and look at the image. Group only if title, description,

image and official UK info agree.

2. Equivalences: IV = I/V = i.v. = Intravenous. IM = I/M = i.m. = Intramuscular. SC = S/C = SubQ = Subcutaneous. Treat them as identical.
3. Status/admin words are NOT part of the name: "out of stock", "prescription only", "POM", "stock item VAT payable", "prescription item zero VAT", "available on request", "training required", "chilled", "new". Strip them from the name and store them in separate columns (stock, rx, vat, ...). They never create a new group.
4. Descriptive words do not create a new product (e.g. "vitamin c b" for Pabrinex: it is always the B+C mix; "high potency"; "injection"; "ampoules").
5. "Concentrate" / "concentration" does NOT indicate the route. Both IV and IM products have strength/concentration. Never infer route from that word, or from any single title word.
6. If the title lacks route/dose/volume, get it from the page description and official reference. If the page contradicts the official UK product, or contradicts itself, do NOT guess: set `needs_human_review = true` and explain.
7. "clone", "replaced for X", "alternative to X", "equivalent": decide from the page and reference whether it is the same UK product under a generic name or a different product. If same product, put it in that variant; if different, keep separate and link with `equivalent_to`. Record the evidence.
8. Bundles/kits ("A + B", "kit", "bundle") are separate items, flagged `is_bundle`.
9. Where one page offers a selector with several routes/packs, create one output row per option.
10. Use the image: if the pack/ampoule image shows a different product, route or strength than the text, flag it.
11. Never merge different models of one brand (e.g. Ultra 2 vs Ultra 3; Teosyal RHA vs Redensity) or lidocaine vs non-lidocaine.

Confidence

Each row gets `confidence`: `confirmed` (page + official source agree), `likely` (page clear, no official source),
`uncertain` (conflict/insufficient info -> needs_human_review).

Pilot first (stop and show me before the full run)

Run on ~60 rows: all Pabrinex rows, plus Juvederm, Teosyal, Botox/Bocouture/Azzalure, Profhilo, needles/cannulas, a few oral medicines.
Known test case: Skin On You "Pabrinex High Potency Concentrate For Injection" - its text

says
"administered intramuscularly" in some places and "intravenously" in one sentence, packaging vague
(5 ml ampoules, 10 per box). Correct output = flagged uncertain/conflict, not a confident guess.
Other Pabrinex titles: Refine "pabrinex" (no route), Aesthetic Pharma "Vitamins B+C Intravenous ... (replaced for Pabrinex)",
Ask Pharmacy "vitamin c im 2ml single ampoule ... pabrinex im clone". Check each from its page.
Note (unverified): a hospital notice says Pabrinex IM was discontinued in the UK; verify against emc/MHRA.

Outputs (in `output/`)

- `products_grouped.csv` : id, supplier, title, url, family_id, family_name, variant_id, variant_label (route / strength / lidocaine / market), pack_id, pack_label (count + unit volume), form, route, rx, vat_note, stock_note, is_bundle, equivalent_to, image_url, evidence (short quote from page + reference url), confidence, needs_human_review, review_reason.
- `reference_catalog.csv` as above.
- `review_queue.csv` : all uncertain rows, sorted by number of suppliers in the family.
- `run_report.md` : counts (rows -> families -> variants -> packs), error counts, top conflicts, rules that needed changing.
  Work supplier by supplier (or platform by platform), commit progress often, keep logs, make every step resumable.
  Do not overwrite cache. Ask me when a rule is ambiguous instead of guessing.